

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 26

FACTORIAL

AIM: To write and execute an 8085 assembly language program to find Factorial in an array and store result in memory.

ALGORITHM

1. Store the given number n at memory location **2050H**.
2. Load n into register **C**.
3. Initialize the result to **1**.
4. Multiply the result by each number from n down to 1.
5. Repeat until the counter becomes zero.
6. Store the factorial result at **2051H**.
7. Stop the program.

CODE:

```
LDA 2050H
```

```
MOV C,A
```

```
MVI A,01H
```

```
MOV B,A
```

```
LOOP: MOV A,B
```

```
MOV D,C
```

```
MVI B,00H
```

```
MUL: ADD B
```

```
DCR D
```

```
JNZ MUL
```

```
MOV B,A
```

```
DCR C
```

```
JNZ LOOP
```

MOV A,B

STA 2051H

HLT

OUTPUT:

The screenshot displays the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window is titled "GNUSim8085 - 8085 Microprocessor Simulator". The interface includes a menu bar (File, Reset, Assembler, Debug, Help) and a toolbar with various icons. The central area shows the assembly code being executed, with line numbers 1 through 17. The code includes instructions like LXI H, 9050, MOV B, M, MVI D, 01H, FACT: CALL MUL, DCR B, JNZ FACT, INX H, MOV M, D, HLT, MUL: MOV E, B, XRA A, ML: ADD D, DCR E, JNZ ML, MOV D, A, RET, and HLT. The left panel shows the Registers section with values for A, BC, DE, HL, PSW, PC, SP, and Int-Reg. The right panel shows the Memory section with a table of addresses and data. The bottom status bar indicates "Simulator: Idle".

Registers

Register	Value
A	00
BC	00 00
DE	00 00
HL	17 71
PSW	00 00
PC	42 1B
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	1
AC	0
P	1
C	0

Assembly Code

```
1 LXI H, 9050
2 MOV B, M
3 MVI D, 01H
4 FACT: CALL MUL
5 DCR B
6 JNZ FACT
7 INX H
8 MOV M, D
9 HLT
10 MUL: MOV E, B
11 XRA A
12 ML: ADD D
13 DCR E
14 JNZ ML
15 MOV D, A
16 RET
17 HLT
```

Memory

Address (Hex)	Address	Data
0802	2050	5
0803	2051	78
0804	2052	0
0805	2053	0
0806	2054	0
0807	2055	0
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0

Assembler Message

```
0 Program assembled successfully
```